discriminative learning $p(y|x)$

generative learning $p(x|y)\,\underbrace{p(y)}_{\text{class priors}}$

Note. $$\arg\max_y P(y|x) = \arg\max_y \frac{P(x|y)\,P(y)}{P(x)}$$

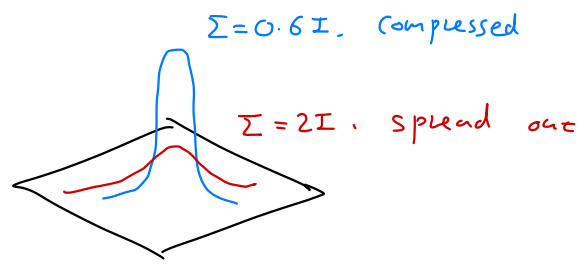
$$= \arg\max_y P(x|y)\,P(y)$$

with $$P(x) = \sum_y P(x|y)\,P(y)$$

Multivariate Normal Distribution

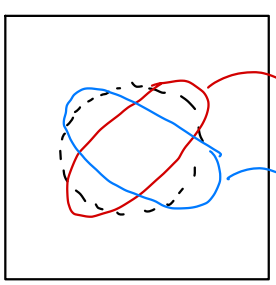
- $$p(x:\mu,\Sigma) = \frac{1}{(2\pi)^{d/2}\,|\Sigma|^{1/2}} \cdot \exp\left(-\tfrac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right)$$

- $$E[X] = \int_X x\,p(x:\mu,\Sigma)\,dx = \mu$$



Note. $$\mathrm{Cov}[Z] = E[(Z-E[Z])(Z-E[Z])^T]$$

$$\mathrm{Cov}[Z] = E[ZZ^T] - E[Z]E[Z]^T$$



$$\Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$$

Gaussian Discriminant Analysis Model (GDA)

$$y \sim \text{Bernoulli}(\phi)$$
$$x \mid y=0 \sim \mathcal{N}(\mu_0, \Sigma)$$
$$x \mid y=1 \sim \mathcal{N}(\mu_1, \Sigma)$$

likelyhood $\mathcal{L} = \prod_{i=1}^{n} p(x^{(i)} \mid y^{(i)})\, p(y^{(i)})$

Apply MLE : $\ell = \sum_{i=1}^{n} \log p(x^{(i)} \mid y^{(i)}) + \sum_{i=1}^{n} \log p(y^{(i)})$

(i) $\frac{\partial \ell}{\partial \phi} = 0$ implies $\phi = \frac{1}{n} \sum_{i=1}^{n} y^{(i)}$

(ii) $\frac{\partial \ell}{\partial \mu_0} = 0$ implies $\mu_0 = \dfrac{\sum_{i=1}^{n} \mathbb{1}\{y^{(i)}=0\}\, x^{(i)}}{\sum_{i=1}^{n} \mathbb{1}\{y^{(i)}=0\}}$

(iii) $\frac{\partial \ell}{\partial \mu_1} = 0$ implies $\mu_1 = \dfrac{\sum_{i=1}^{n} \mathbb{1}\{y^{(i)}=1\}\, x^{(i)}}{\sum_{i=1}^{n} \mathbb{1}\{y^{(i)}=1\}}$

(iv) $\frac{\partial \ell}{\partial \Sigma} = 0$ implies $\Sigma = \frac{1}{n} \sum_{i=1}^{n} (x^{(i)} - \mu_{y^{(i)}})(x^{(i)} - \mu_{y^{(i)}})^T$

GDA with shared Σ $\Rightarrow$ logistic Regression

Note. Logistic Regression

$$p(y=1 \mid x) = \frac{1}{1+e^{-z}}$$

with $z = w^T x + b$

- $p(y=1 \mid x) = 0.5$
- $p(x \mid y=0)\, p(y=0) = p(x \mid y=1)\, p(y=1)$
- $p(y=1 \mid x) = \dfrac{1}{1 + \dfrac{p(x \mid y=0)\, p(y=0)}{p(x \mid y=1)\, p(y=1)}}$
- $$\ln\left(\frac{p(x \mid y=1)\, p(y=1)}{p(x \mid y=0)\, p(y=0)}\right) = \begin{aligned} &+ \tfrac{1}{2}(x-\mu_0)^T \Sigma^{-1} (x-\mu_0) \\ &- \tfrac{1}{2}(x-\mu_1)^T \Sigma^{-1} (x-\mu_1) \end{aligned} + \ln\left(\frac{\phi}{1-\phi}\right)$$

$$= (\mu_1^T \Sigma^{-1} - \mu_0^T \Sigma^{-1})\, x$$
$$- \tfrac{1}{2} \mu_1^T \Sigma^{-1} \mu_1 + \tfrac{1}{2} \mu_0^T \Sigma^{-1} \mu_0 + \ln\left(\frac{\phi}{1-\phi}\right)$$

logistic Regression $\not\Rightarrow$ GCD (Shared Σ)

- GDA makes stronger assumptions than Logistic Regression. (fits well when assumption holds)

- LR is more robust.

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if $p(x|y)$ is Gaussian with shared Σ

then GDA is "asymptotically correct"

i.e large training set $\Rightarrow$ no algorithm better at estimating $p(y|x)$

Note. in small training set GDA still performs well!

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if $x \mid y=0 \sim \text{Poisson}(\lambda_0)$

$x \mid y=1 \sim \text{Poisson}(\lambda_1)$

$\Rightarrow P(y|x)$ is logistic (LR)